



7261 - JWST Simulations of Molecular Outflows around Star-forming Galaxies

Cycle: 4, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

Theoretical studies of warm (100-1000 K) and hot (>1000 K) molecular superwind outflows traversing from the ISM through the disk-halo interface and into the circumgalactic medium (CGM) are substantially under-developed considering JWST's groundbreaking capability to spectrally resolve and spatially map numerous H₂ transitions. To properly simulate JWST observations of these outflows, one needs 1) ultra-high-resolution hydrodynamic simulations of galaxies with 2) sophisticated treatment of stellar feedback including supernovae, stellar winds, and radiation pressure while 3) self-consistently tracking the creation and propagation of molecular species and finally 4) mocking realistic observations using 3-dimensional radiative transfer. We propose hydrodynamic simulations of galaxies using the FIRE-2 code integrated with the novel CHIMES (CHemistry of the Interstellar Medium and Extragalactic Sources) module that tracks the non-equilibrium molecular chemistry and thermodynamics to create synthetic MIRI and NIRSpec spectra and Integral Field Unit (IFU) data cubes. A state-of-the-art multi-element dust depletion and shielding model makes it possible to simulate molecular outflows traversing from dense, HII regions in the ISM to the diffuse multi-phase CGM for the first time. We will diagnose how the underlying physical properties of superwinds (mass-loading, velocity, & total power) are observed across a variety of star-forming galaxies, how outflow properties scale with star formation rate surface density, and create a public data repository of synthetic MIRI, NIRSpec, and NIRCам observations to interpret existing observations and inform future endeavors.

OBSERVING DESCRIPTION